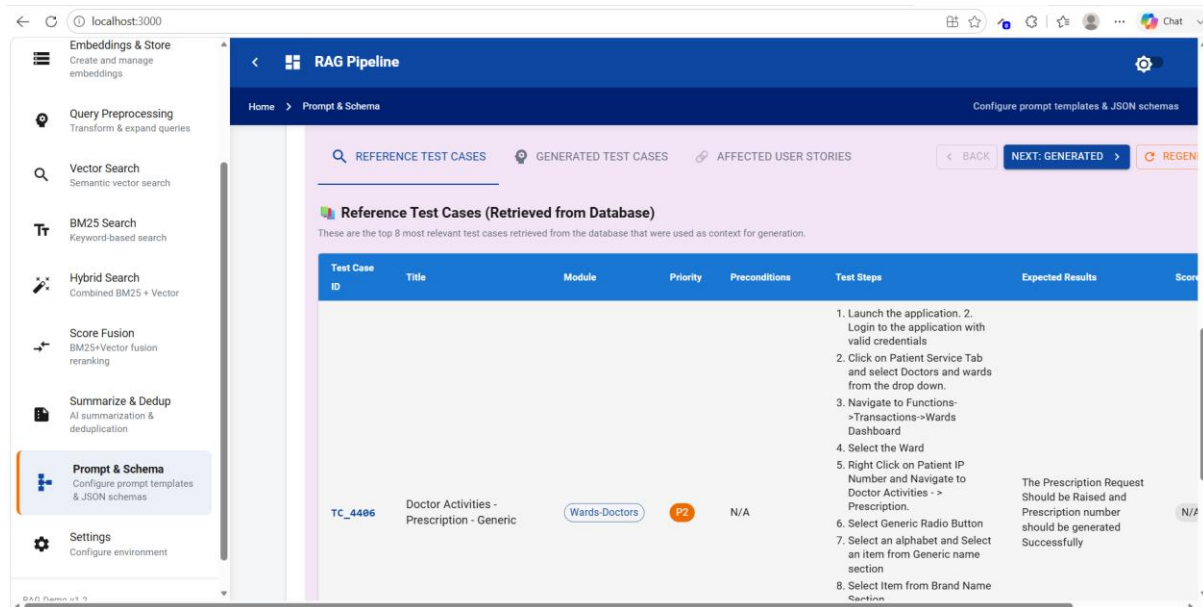


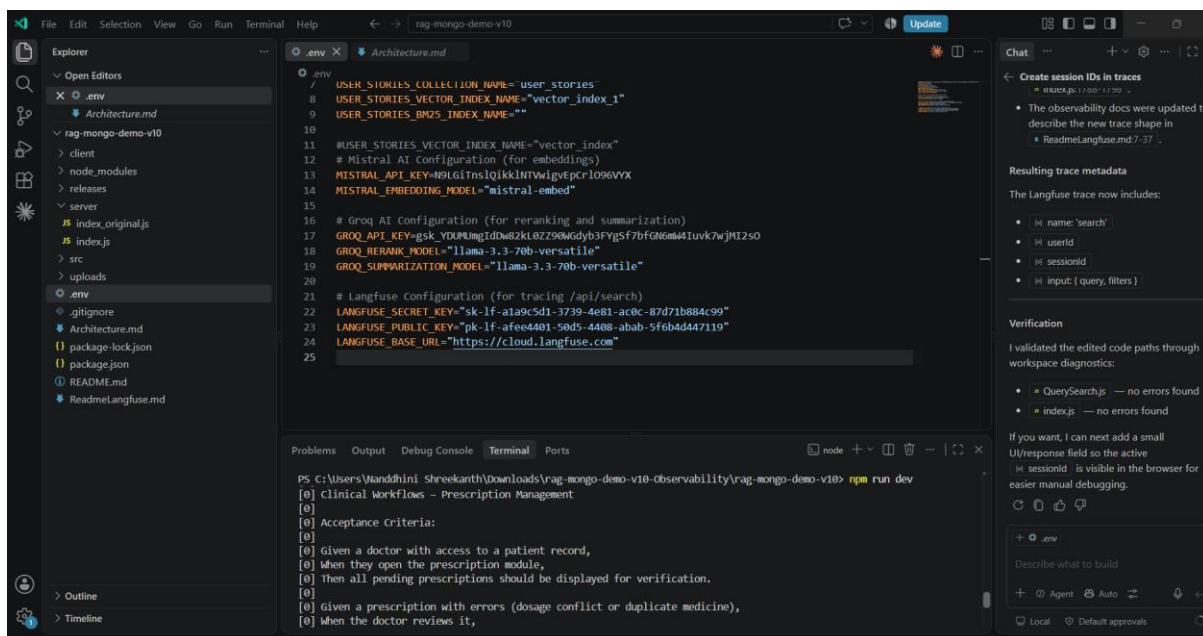
Observability: -

Vector Search

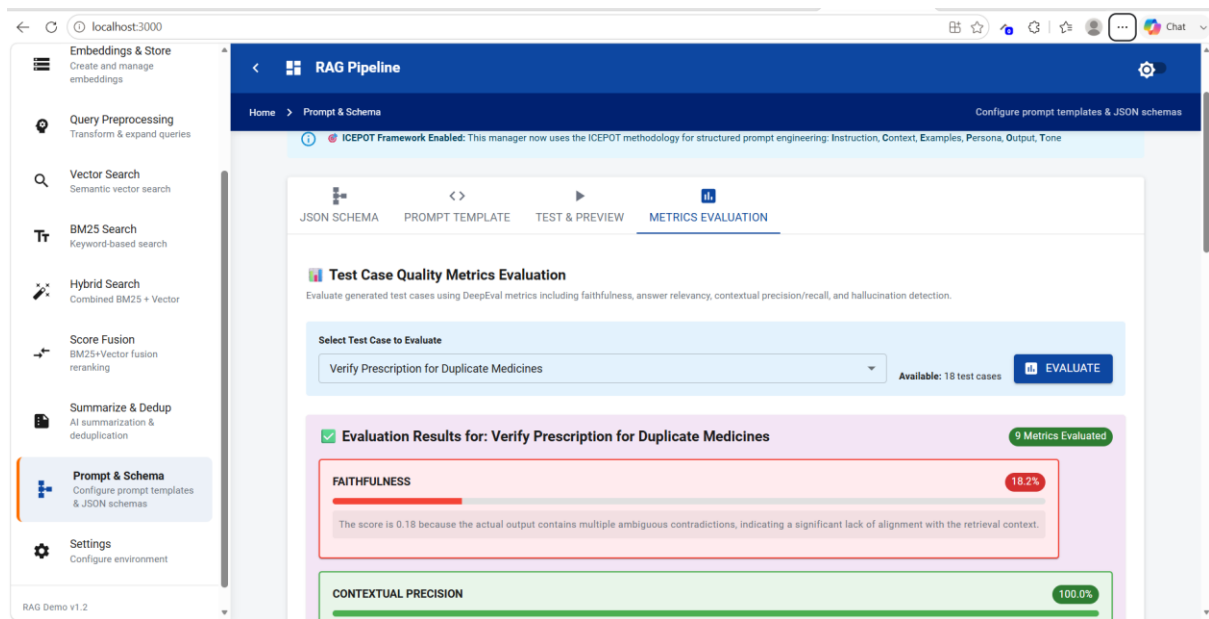
Validated the test and Preview by giving the User story



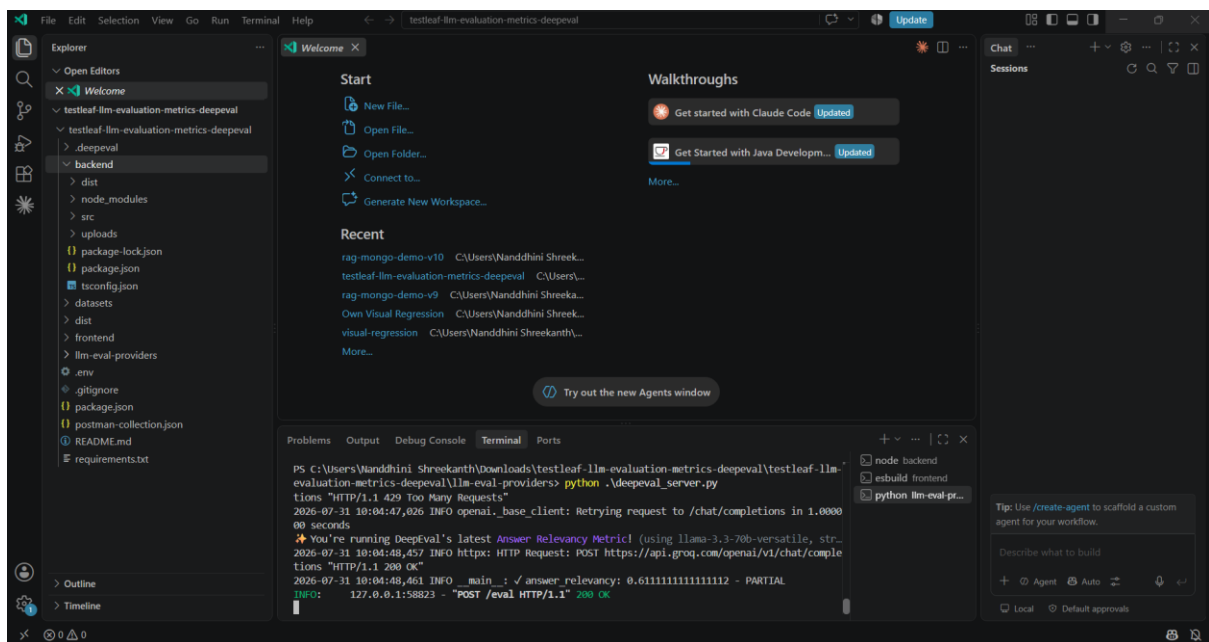
Validated the Logs in VS code



Validated Metrics Evaluation



Validated the logs in VS code for DeepEval



Validated the results in Vector search

The screenshot displays the RAG Pipeline Vector Search interface. The left sidebar contains navigation options: embeddings, Query Preprocessing, Vector Search (selected), BM25 Search, Hybrid Search, Score Fusion, Summarize & Dedup, Prompt & Schema, and Settings. The main area shows the 'Query Search' section with a search query input field containing 'Share Diagnostic Reports with Patients via Email'. Below the input field, there are filters for 'Advanced Metadata Filters' and 'Results Limit' set to '5 Results'. A 'SEARCH' button is present. The search results are displayed in a table with columns: Test Case, Similarity, Description, and Source. The results are ranked by semantic similarity. The first result is 'Speciality EMR & Care Plans' with a similarity of 89.4%. The second result is 'Investigation report of patients' with a similarity of 89.3%. The third result is 'Reports - Clinical Reports - View Report' with a similarity of 89.3%. The fourth result is 'Reports - Clinical Reports - View Report' with a similarity of 89.3%. The fifth result is 'Doctors-Wards' with a similarity of 89.3%. The interface also shows a 'Detailed View' section for the selected result, 'Speciality EMR & Care Plans', with a score of 0.8938 and a source of 'IP Digital'.

Query Search

Search through your test cases using semantic similarity with AI embeddings.

Search Query
Enter your search terms and find similar test cases

Enter your search query
Share Diagnostic Reports with Patients via Email

Use descriptive terms related to test functionality

Advanced Metadata Filters Results Limit: 5 Results

Search Query: "Share Diagnostic Reports with Patients via Email" 5 results
Performance: (\$0.000001 cost) (12 tokens)

Search Results

5 results ranked by semantic similarity

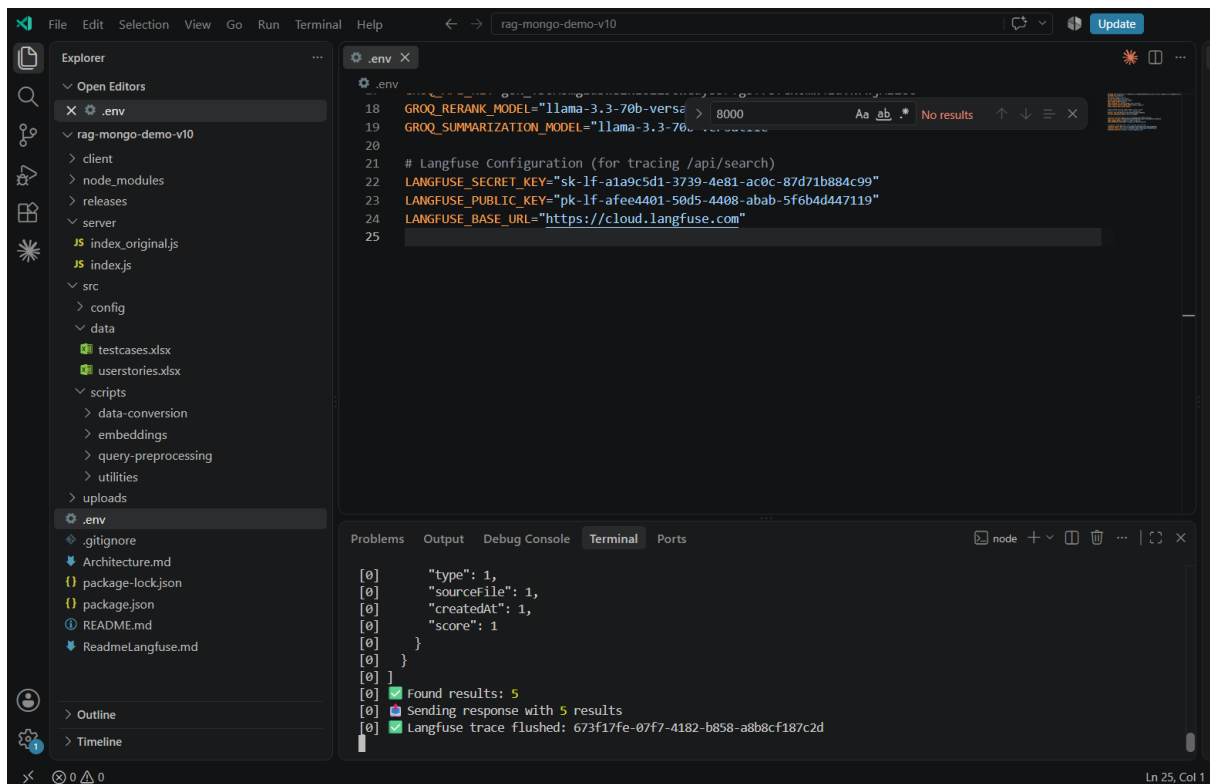
Test Case	Similarity	Description	Source
Speciality EMR & Care Plans	89.4%	To view reports	☐
IP Digital			
Investigation report of patients	89.3%	Investigation report of patients	☐
IP digital			
Reports - Clinical Reports - View Report	89.3%	Verify user able to view Clinical Report	☐
Doctors-Wards			
Reports - Clinical Reports - View Report	89.3%	Verify user able to view Clinical Report	☐
Doctors-Wards			

Rows per page: 10 1-5 of 5

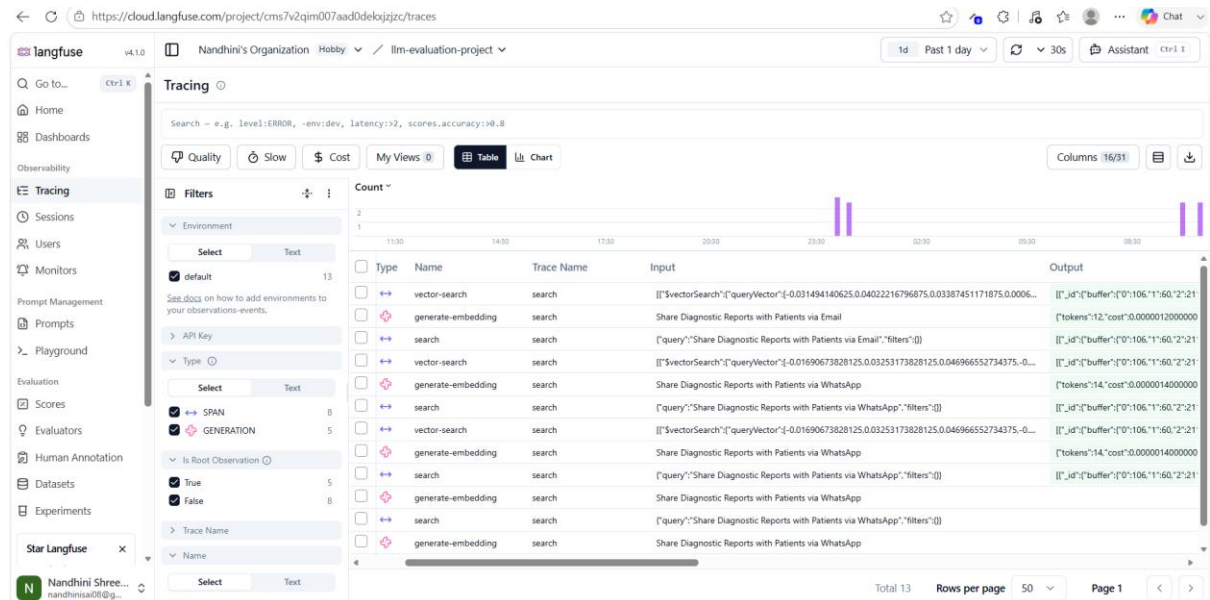
Detailed View

Speciality EMR & Care Plans Score: 0.8938 IP Digital

Validated the log for Vector search



Tracing >> Vector search



Users

Go to...

Home

Dashboards

Observability

Tracing

Sessions

Users

Monitors

Prompt Management

Prompts

Playground

Evaluation

Scores

Evaluators

Human Annotation

Datasets

Experiments

Star Langfuse

Nandhini Shree...

langfuse

v4.1.0

Nandhini's Organization

Hobby

llm-evaluation-project

1d

Past 1 day

Assistant

Ctrl I

Users

Search (User ID)

Env default

Filters

User ID	Environment	First Event	Last Event	Total Events	Total Tokens	Total Cost
09564039-d09c-4495-81ee-1f59d47df02c	default	31/7/2026, 12:29:25 am	31/7/2026, 10:41:35 am	13	40	\$0.00

Rows per page 50Page 1 of 1

Go to...

Home

Dashboards

Observability

Tracing

Sessions

Users

Monitors

Prompt Management

Prompts

Playground

Evaluation

Scores

Evaluators

Human Annotation

Datasets

Experiments

Star Langfuse

Nandhini Shree...

langfuse

v4.1.0

Nandhini's Organization

Hobby

llm-evaluation-project

Users

Assistant

Ctrl I

User 09564039-d09c-4495-81ee-1f59d47df02c

Dashboard

Observations: 13Traces: 5Total Tokens: 40Total Cost: \$0.00Active: 31/7/2026, 12:29:25 am - 31/7/2026, 10:41:35 am

TracesSessionsScores

QualitySlowCostMy ViewsSearchIDs / Names1dPast 1 day30sColumns 16/31

Filters	Start Time	Type	Name	Trace Name	Input	Output
Environment	2026-07-31 10:41:35	vector-search	search		[{"vectorSearch":{"queryVector":[-0.031494140625,0.04022216796875,0.03387451171875,0.0006...	[{"_id":"...
default	2026-07-31 10:41:33	generate-embedding	search		Share Diagnostic Reports with Patients via Email	[{"tokens...
	2026-07-31 10:41:31	search	search		("query":"Share Diagnostic Reports with Patients via Email","filters":[])	[{"_id":"...
	2026-07-31 10:15:04	vector-search	search		[{"vectorSearch":{"queryVector":[-0.01690673828125,0.03253173828125,0.046966552734375,-0...	[{"_id":"...
	2026-07-31 10:15:04	generate-embedding	search		Share Diagnostic Reports with Patients via WhatsApp	[{"tokens...
	2026-07-31 10:15:02	search	search		("query":"Share Diagnostic Reports with Patients via WhatsApp","filters":[])	[{"_id":"...
	2026-07-31 00:40:28	vector-search	search		[{"vectorSearch":{"queryVector":[-0.01690673828125,0.03253173828125,0.046966552734375,-0...	[{"_id":"...
	2026-07-31 00:40:27	generate-embedding	search		Share Diagnostic Reports with Patients via WhatsApp	[{"tokens...
	2026-07-31 00:40:26	search	search		("query":"Share Diagnostic Reports with Patients via WhatsApp","filters":[])	Share Diagnostic Reports with Patients via WhatsApp
	2026-07-31 00:29:55	generate-embedding	search		Share Diagnostic Reports with Patients via WhatsApp	
	2026-07-31 00:29:53	search	search		("query":"Share Diagnostic Reports with Patients via WhatsApp","filters":[])	
	2026-07-31 00:29:27	generate-embedding	search		Share Diagnostic Reports with Patients via WhatsApp	
	2026-07-31 00:29:25	search	search		("query":"Share Diagnostic Reports with Patients via WhatsApp","filters":[])	

Total 13Rows per page 50Page 1

Sessions: -

ID	Created At	Duration	Environment	Scores	User IDs	Traces
de22f0f6-cef9-4e15-8d43-94a281fa...	2026-07-31 10:41:31	4.00s	default		09564039-d09c-4495-81ee-1f59d47d02c	1
6b4c37-67f3-4613-98db-0a56a0a...	2026-07-31 10:15:02	2.00s	default		09564039-d09c-4495-81ee-1f59d47d02c	1

Path	Value
tokens	12
cost	0.000001200000000000000002

Formatted JSON

```
{  "input": "Share Diagnostic Reports with Patients via Email",  "output": {    "tokens": 12,    "cost": 0.000001200000000000000002  },  "vector_search": {    "span": "31/7/2026, 10:41:35 am",    "limit": 5,    "project": "vector-search"  },  "formatted_json": {    "0": {      "vector_search": {        "span": "31/7/2026, 10:41:35 am",        "limit": 5,        "project": "vector-search"      }    },    "1": {      "vector_search": {        "span": "31/7/2026, 10:41:35 am",        "limit": 5,        "project": "vector-search"      }    },    "2": {      "vector_search": {        "span": "31/7/2026, 10:41:35 am",        "limit": 5,        "project": "vector-search"      }    },    "3": {      "vector_search": {        "span": "31/7/2026, 10:41:35 am",        "limit": 5,        "project": "vector-search"      }    },    "project": {      "vector_search": {        "span": "31/7/2026, 10:41:35 am",        "limit": 5,        "project": "vector-search"      }    }  }  }
```

Prompt to implement the BM25 and Hybrid search

Prompt: Implement Langfuse Tracing for BM25 Search and Hybrid Search

The project already has Langfuse tracing implemented for the **Vector Search** endpoint (POST /api/search).

Your task is to implement **the exact same tracing architecture** for:

- POST /api/search/bm25
- POST /api/search/hybrid

The implementation should follow the existing Vector Search implementation as closely as possible. Do **not** redesign or refactor the tracing architecture.

Goal

Maintain one consistent Langfuse tracing architecture across all search APIs.

Each request should produce one Langfuse trace with child spans representing each major search operation.

Reuse the existing Langfuse singleton client already initialized in server/index.js.

Do not introduce helper wrappers or abstraction layers.

Existing implementation

The existing endpoint /api/search already creates:

trace: "search"

```
|
|
|— generation: "generate-embedding"
|
|
└— span: "vector-search"
```

Follow the exact same coding style, lifecycle, naming conventions, and flush strategy.

Part 1 — BM25 Search

Implement Langfuse tracing for:

POST /api/search/bm25

Trace

Create one trace per request.

trace: "bm25-search"

Input:

```
{
  "query": "...",
  "filters": { ... },
  "limit": 10
}
```

Output:

The exact results array returned to the client.

No additional serialization.

Child span

Create one child span.

span: "bm25-search"

This span must wrap only:

```
collection.aggregate(pipeline).toArray()
```

Span input

Record the aggregation pipeline exactly as constructed.

Do not reshape it.

Include:

- Atlas Search stage
 - Filters
 - Projection
 - Limit
 - Sort (if present)
-

Span output

Record:

- returned documents
- searchScore
- result count

Use the same objects already returned by MongoDB.

No additional processing.

Flush

Exactly the same as Vector Search.

Success path:

```
await langfuse.flushAsync();
```


before

res.json(...)

Failure path:

await langfuse.flushAsync();

inside catch.

Part 2 — Hybrid Search

Implement Langfuse tracing for:

POST /api/search/hybrid

Hybrid Search performs multiple independent operations.

Each major operation should be represented as its own child span.

Trace

Create one trace:

trace: "hybrid-search"

Input:

```
{  
  "query": "...",  
  "filters": { ... },  
  "limit": 10  
}
```

Output:

The exact response returned to the client.

No duplicate serialization.

Child span 1

generation: "generate-embedding"

Wrap only:

generateEmbedding(query)

Record:

Model:

mistral-embed

Input:

query

Output:

```
{  
  tokens,  
  cost  
}
```

Reuse the existing token usage and cost calculations.

Do not introduce new pricing logic.

Child span 2

span: "vector-search"

Wrap only:

`collection.aggregate(vectorPipeline).toArray()`

Input:

Vector aggregation pipeline.

Output:

Returned vector results.

Include scores exactly as returned.

Child span 3

span: "bm25-search"

Wrap only:

`collection.aggregate(bm25Pipeline).toArray()`

Input:

BM25 aggregation pipeline.

Output:

Returned BM25 results.

Include Atlas Search scores.

Child span 4

Create a final span representing Hybrid fusion.

span: "merge-results"

Wrap only the logic that combines the Vector Search and BM25 Search results into the final ranked response.

Input:

```
{  
  "vectorResults": [...],  
  "bm25Results": [...]  
}
```

Output:

```
{  
  "mergedResults": [...]  
}
```

Record the final ranked results exactly as returned by the merge logic.

Do not modify the ranking algorithm.

This span should measure only the merge/ranking phase.

Expected Hybrid Trace

trace: "hybrid-search"

input:

query

filters

limit

|

|— generation: "generate-embedding"

|

|— span: "vector-search"

|

|— span: "bm25-search"

|

|— span: "merge-results"

|

output:

final results

Client

Reuse the existing singleton Langfuse client.

Do NOT:

- create another Langfuse instance
- create helper classes
- move tracing into middleware

Use the same implementation pattern already used by /api/search.

Flush strategy

Exactly match the existing implementation.

Every endpoint must call

`await langfuse.flushAsync();`

Success:

Immediately before

`res.json(...)`

Failure:

Inside catch.

Error handling

If any search phase throws:

- Record the exception on the active trace.
- End the active span.
- Flush Langfuse.
- Preserve the existing API behavior.

- Do not change response formats.
-

Out of scope

Do NOT implement:

- Generic tracing helpers
- Middleware-based tracing
- Sampling
- Feature flags
- Configuration switches
- Additional metrics
- Refactoring of existing search logic
- Changes to ranking algorithms
- Changes to MongoDB queries
- New abstractions

Follow the current Vector Search implementation exactly.

Verification

BM25

Verify that one trace appears:

bm25-search

with:

- Trace input
 - BM25 child span
 - Aggregation pipeline
 - Search scores
 - Final response
-

Hybrid

Verify that one trace appears:

hybrid-search

with four nested children:

generate-embedding

vector-search

bm25-search

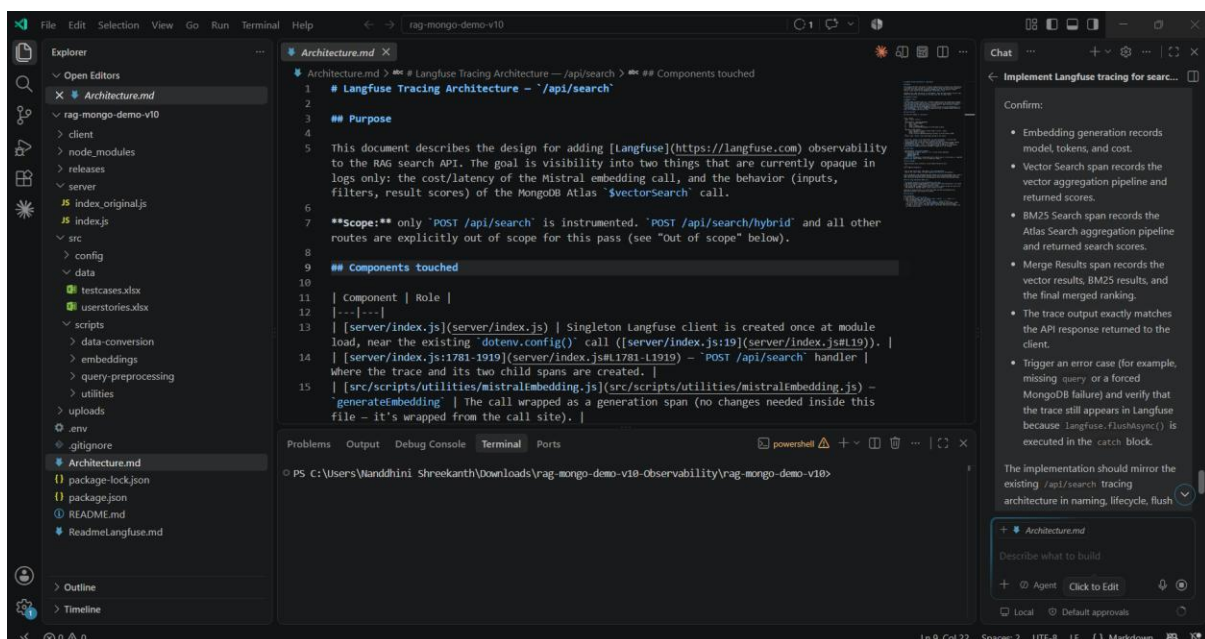
merge-results

Confirm:

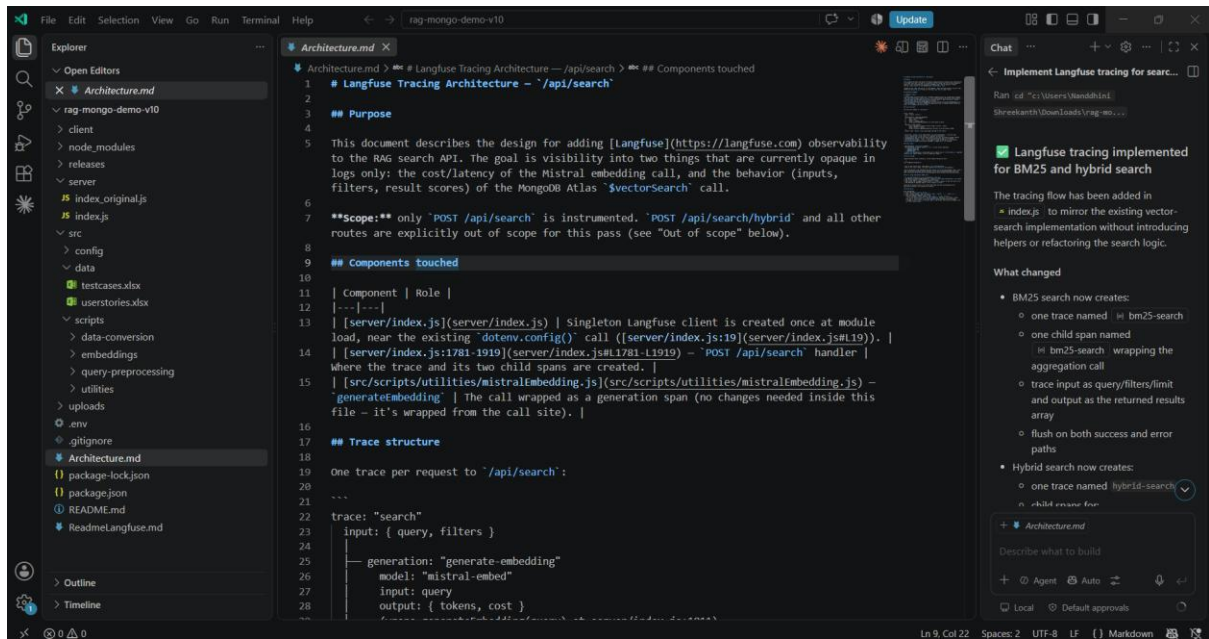
- Embedding generation records model, tokens, and cost.
- Vector Search span records the vector aggregation pipeline and returned scores.
- BM25 Search span records the Atlas Search aggregation pipeline and returned search scores.
- Merge Results span records the vector results, BM25 results, and the final merged ranking.
- The trace output exactly matches the API response returned to the client.
- Trigger an error case (for example, missing query or a forced MongoDB failure) and verify that the trace still appears in Langfuse because `langfuse.flushAsync()` is executed in the catch block.

The implementation should mirror the existing `/api/search` tracing architecture in naming, lifecycle, flush strategy, and overall coding style, ensuring consistent observability across Vector Search, BM25 Search, and Hybrid Search.

Given the Prompt for Vibe coding to implement the BM25 search and Hybrid search

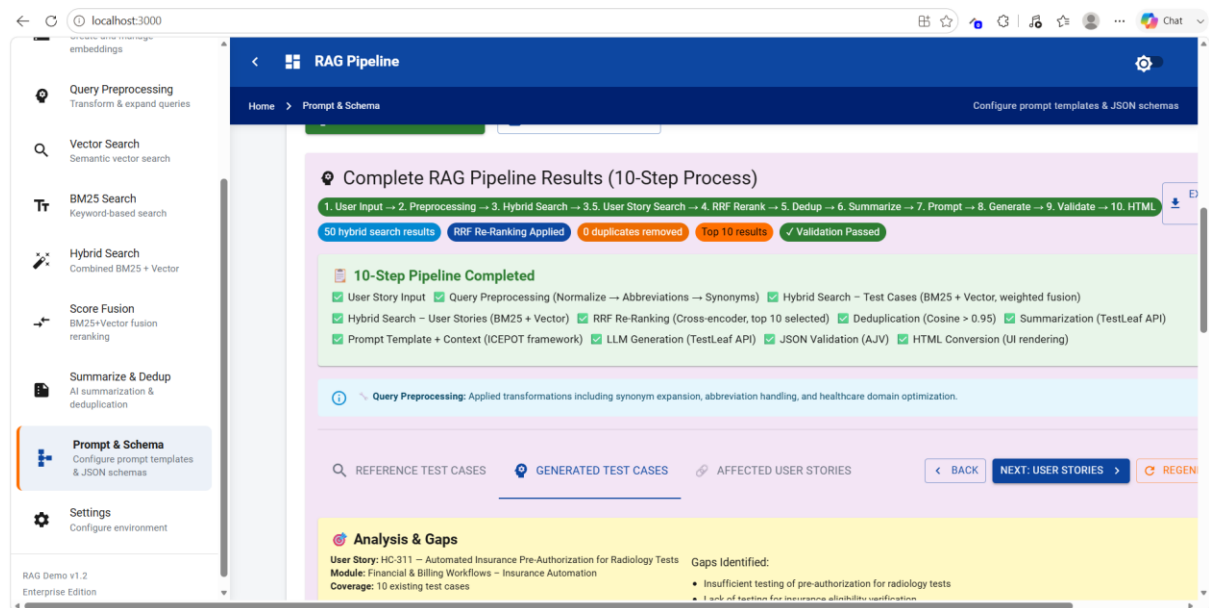


Implemented the BM25 search and Hybrid search using Vibe coding

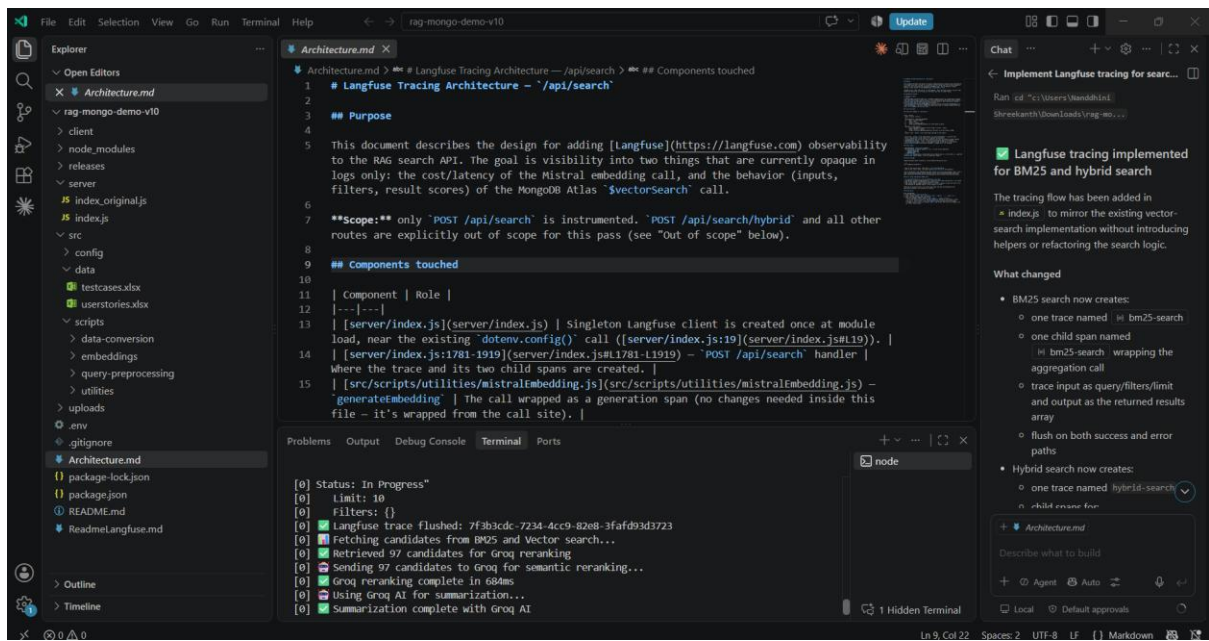


BM25 Search

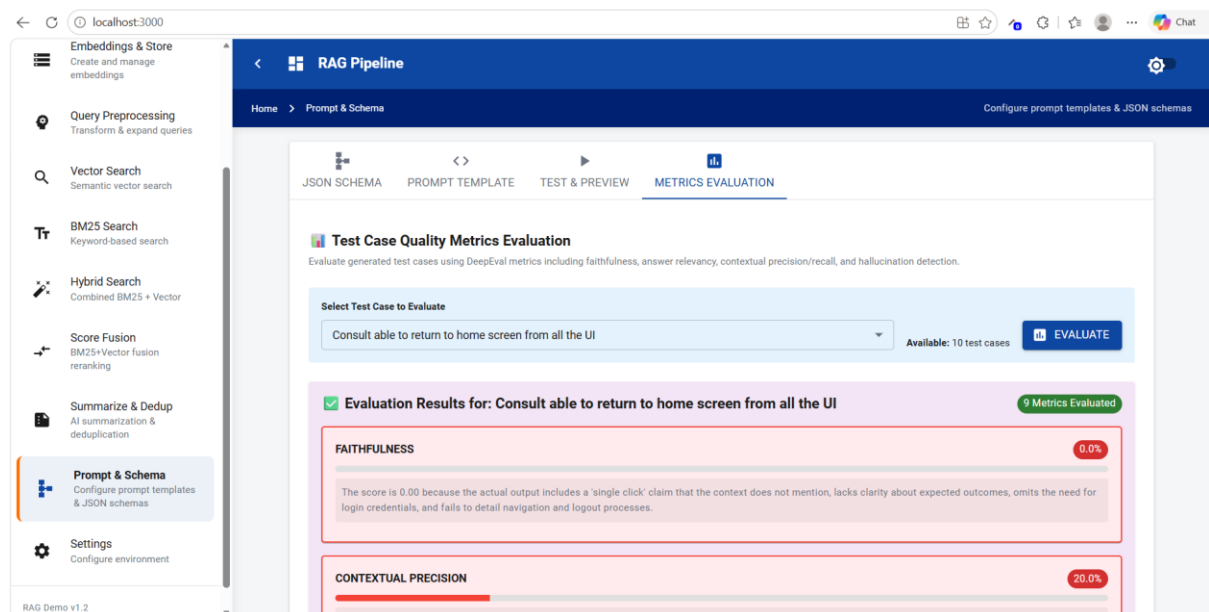
Validated the RAG pipeline



Validated the Logs in VS code



Validated the Metrics Evaluation



Validated the logs in VS code for DeepEval

The screenshot shows the VS Code interface with the following components:

- Explorer:** Shows the file structure of the `testleaf-llm-evaluation-metrics-deepeval` project, including `.env`, `testleaf-llm-evaluation-metrics-deepeval`, `testleaf-llm-evaluation-metrics-deepeval`, `enterprise_eval_dataset.xlsx`, `testcasagen_agent_eval_dataset.xlsx`, `data`, `frontend`, `llm-eval-providers`, `.env`, `.gitignore`, `package.json`, `postman-collection.json`, `README.md`, and `requirements.txt`.
- Code Editor:** Displays the `.env` file with the following configuration:

```
1 # DeepEval Server Configuration
2 # Host/port used by the FastAPI sidecar. Keep this aligned with the backend's
3 # DEEPEVAL_URL when using the default local setup.
4 PORT=8001
5 # LLM Evaluation Model
6 # Options: gpt-4, gpt-4o, gpt-4o-mini, gpt-3.5-turbo, llama-3.3-70b-versatile, etc.
7 # Default: llama-3.3-70b-versatile
8 # EVAL_MODEL=llama-3.3-70b-versatile
9 OPENROUTER_API_KEY=sk-or-v1-50888227171c7296728f4659e39a79d4e5478f1c03f7952f1c4dee2adde6b58
10 OPENROUTER_BASE_URL=https://openrouter.ai/api/v1
11 OPENROUTER_MODEL=openai/gpt-4o-mini
12 # EVAL_MODEL=gpt-4o-mini
13 # API Keys (choose based on EVAL_MODEL)
14 # For Groq models (llama-*, mixtral-*, etc.)
15 GROQ_API_KEY=gsk_oZfESYteq7xVvY4fL8JWgdyb3FYjEFZHZSLZQ6476HJM4q18U
16 # For OpenAI models (gpt-*)
17 OPENAI_API_KEY=
18 # IDK Handling Configuration for Faithfulness Metric
19 # How to treat "idk" in faithfulness evaluation:
20 # - "yes": Treat ambiguous claims as supported (lenient scoring, higher scores)
21 # - "no": Treat ambiguous claims as unsupported (strict scoring, lower scores)
22 IDK_HANDLING=no
23
```
- Terminal:** Shows the output of the `python .\deepeval_server.py` command. The output indicates that the server is running successfully and that the evaluation results are being processed. The output includes the following information:

```
PS C:\Users\Nandhini Shreekanth\Downloads\testleaf-llm-evaluation-metrics-deepeval\testleaf-llm-evaluation-metrics-deepeval> python .\deepeval_server.py
2026-07-31 15:14:44,045 INFO http: HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
You're running DeepEval's latest Answer Relevancy Metric! (using openai/gpt-4o-mini, strict=False)
2026-07-31 15:14:45,980 INFO http: HTTP Request: POST https://openrouter.ai/api/v1/chat/completions "HTTP/1.1 200 OK"
2026-07-31 15:14:46,597 INFO _main_: ✓ answer_relevancy: 0.0 - NOT_RELEVANT
INFO: 127.0.0.1:55576 - "POST /eval HTTP/1.1" 200 OK
```
- Chat:** Shows a chat window with the following messages:
 - uses OPENROUTER_BASE_URL
 - optionally honors OPENROUTER_MODEL
 - MetricEvaluator now accepts an optional base_url and passes it through to GroqModel

Validated the results in BM25 search

The screenshot shows the RAG Pipeline interface with the following components:

- Left Sidebar:** Contains a list of search methods: `Convert to JSON`, `Embeddings & Store`, `Query Preprocessing`, `Vector Search`, `BM25 Search` (selected), `Hybrid Search`, `Score Fusion`, `Summarize & Dedup`, `Prompt & Schema`, and `Settings`.
- Main Content Area:** Displays the `BM25 Keyword Search` results. The search query is `Download claim form`, and the results are 10. The search type is `bm25`. The results are as follows:

Result ID	Score	Text
#1 TC_1079	Score: 13.94	Fill form Module: Digital OP P4 Risk: Low Manual Description: Verify filled form Steps: 1. Launch the application 2. Login with credentials 3. Mark as 'N' 4. Check in on patient dropdown 5. Enter allergy, if new patient. 6. Click on Invoice form. 7. Click 'Yes' for submitting the details. 8. Search the form and enter details from Search and attach tab. 9. Save the details Expected Results: Filled form should be available in 'View attached form tab' Created: 25/6/2026
#2 TC_872	Score: 12.31	



Tracing >> BM25 search

The screenshot shows the Langfuse Tracing interface for a project named "Nandhini's Organization". The left sidebar contains navigation options: Home, Dashboards, Observability, Tracing (selected), Sessions, Users, Monitors, Prompt Management, Prompts, Playground, Evaluation, Scores, Evaluators, Human Annotation, Datasets, and Experiments. The main area displays a list of traces for the "BM25 search" type. The table has columns: Start Time, Type, Name, Trace Name, and Input. The traces are sorted by Start Time. A bar chart at the top shows the distribution of traces over time. The bottom of the interface shows pagination information: Total 50, Rows per page 50, Page 1.

Start Time	Type	Name	Trace Name	Input
2026-07-31 15:16:42	bm25-search	bm25-search	bm25-search	[{"search":{"index":"default","text":{"query":"Download claim form","path":{"id","title","description...","doc
2026-07-31 15:16:40	bm25-search	bm25-search	bm25-search	[{"query":"Download claim form","filters":{"limit":10}
2026-07-31 14:06:19	bm25-search	hybrid-search	hybrid-search	[{"search":{"index":"default","text":{"query":"User Story: HC-311 — Automated Insurance Pre-Au...
2026-07-31 14:46:23	bm25-search	hybrid-search	hybrid-search	[{"search":{"index":"default","text":{"query":"User Story: HC-311 — Automated Insurance Pre-Au...
2026-07-31 14:42:46	bm25-search	hybrid-search	hybrid-search	[{"search":{"index":"default","text":{"query":"User Story: HC-311 — Automated Insurance Pre-Au...
2026-07-31 14:07:54	bm25-search	hybrid-search	hybrid-search	[{"search":{"index":"default","text":{"query":"User Story: HC-311 — Automated Insurance Pre-Au...
2026-07-31 14:27:40	bm25-search	hybrid-search	hybrid-search	[{"search":{"index":"default","text":{"query":"User Story: HC-311 — Automated Insurance Pre-Au...
2026-07-31 14:23:24	bm25-search	hybrid-search	hybrid-search	[{"search":{"index":"default","text":{"query":"User Story: HC-311 — Automated Insurance Pre-Au...
2026-07-31 11:06:43	bm25-search	hybrid-search	hybrid-search	[{"search":{"index":"default","text":{"query":"User Story: HC-311 — Automated Insurance Pre-Au...
2026-07-31 14:23:24	generate-embedding	hybrid-search	hybrid-search	User Story: HC-311 — Automated Insurance Pre-Authorization for Radiology Tests Summary: As ...
2026-07-31 14:46:23	generate-embedding	hybrid-search	hybrid-search	User Story: HC-311 — Automated Insurance Pre-Authorization for Radiology Tests Summary: As ...
2026-07-31 14:06:19	generate-embedding	hybrid-search	hybrid-search	User Story: HC-311 — Automated Insurance Pre-Authorization for Radiology Tests Summary: As ...

Users

The screenshot shows the Langfuse Users interface for a project named "Nandhini's Organization". The left sidebar contains navigation options: Home, Dashboards, Observability, Tracing, Sessions, Users (selected), Monitors, Prompt Management, Prompts, Playground, Evaluation, Scores, Evaluators, Human Annotation, Datasets, and Experiments. The main area displays a list of users for the "BM25 search" type. The table has columns: Start Time, Type, Name, Trace Name, and Input. The users are sorted by Start Time. A bar chart at the top shows the distribution of users over time. The bottom of the interface shows pagination information: Total 4, Rows per page 50, Page 1.

Start Time	Type	Name	Trace Name	Input
2026-07-31 16:14:56	bm25-search	bm25-search	bm25-search	[{"search":{"index":"default","text":{"query":"Share Diagnostic Reports with Patients via W
2026-07-31 16:14:55	bm25-search	bm25-search	bm25-search	[{"query":"Share Diagnostic Reports with Patients via WhatsApp","filters":{"limit":10}
2026-07-31 16:14:47	bm25-search	bm25-search	bm25-search	[{"search":{"index":"default","text":{"query":"Download claim form","path":{"id","title","de
2026-07-31 16:14:45	bm25-search	bm25-search	bm25-search	[{"query":"Download claim form","filters":{"limit":10}

https://cloud.langfuse.com/project/cms7v2qim007aad0dekojzjc/users/%3A%3A1?peek=t-10a18bd6-b81d-43ce-bc0b-f8cf17bc5ca&observation=t-10a18bd6-b81d-43ce-bc0b-f8cf17bc5ca

langfuse v4.2.0

Trace bm25-search: 10a18bd6-b81d-43ce-bc0b-f8cf17bc5ca

Search

bm25-search 1.82s

bm25-search 0.00s

2026-07-31 16:14:55.685

Session: t-1785494685685 User ID: :1 Env: default

Preview Scores Log View Formatted JSON

Input

Path	Value
query	"Download claim form"
filters	empty object
limit	10

Output

Path	Value
> 0	12 items
> 1	12 items
> 2	12 items
> 3	12 items
> 4	12 items
> 5	12 items
> 6	12 items
> 7	12 items
> 8	12 items
> 9	12 items

Corrected Output

JSON

Sessions

https://cloud.langfuse.com/project/cms7v2qim007aad0dekojzjc/sessions

langfuse v4.2.0

Nandhini's Organization Hobby / ilm-evaluation-project

1d Past 1 day Assistant Ctrl+I

Sessions

My Views 0

Columns 10/16

Filters

Environment

Select Text

default

See docs on how to add environments to your sessions.

Session ID

User IDs All

Trace Tags

Metadata

Bookmarked

Bookmarked

Not bookmarked

Session Duration

Traces Count

Input Tokens

Output Tokens

Total Tokens

Input Cost

ID	Created At	Duration	Environment	Scores	User IDs
t-1785494695405	2026-07-31 16:14:55	1.00s	default		:1
t-1785494685685	2026-07-31 16:14:45	2.00s	default		:1
da22f0f6-cef9-4e15-8d43-94a28f1a...	2026-07-31 10:41:31	4.00s	default		09564039-d09c-4495-4
6b4cbc37-67f3-4613-98db-0a56a0a...	2026-07-31 10:15:02	2.00s	default		09564039-d09c-4495-4

Rows per page 50 Page 1 of 1

Screenshot of the Langfuse web interface showing a session view for a project named "Nandhini's Organization". The session ID is "1-1785494685685". The interface displays input and output data for a search query.

Input:

Path	Value
query	"Download claim form"
filters	empty object
limit	10

Output:

Path	Value
> 0	12 items
> 1	12 items
> 2	12 items
> 3	12 items
> 4	12 items
> 5	12 items
> 6	12 items
> 7	12 items
> 8	12 items
> 9	12 items

Scores:

The interface also shows a sidebar with navigation options like Home, Dashboards, Observability, Tracing, Sessions, Users, Monitors, Prompt Management, Prompts, Playground, Evaluation, Scores, Evaluators, Human Annotation, Datasets, and Experiments. The bottom status bar indicates the session was run on 31/7/2026 at 4:14:47 pm.

Hybrid Search

Screenshot of the RAG Pipeline web interface showing the Hybrid Search configuration. The interface allows users to adjust search weights between BM25 and Vector search.

Hybrid Search (BM25 + Vector)

Combines BM25 keyword matching with semantic vector search for best results. Adjust weights to control the balance.

Search Query: Book appointment with specialist doctor

Results Limit: 10

Search Weights:

- BM25 Weight: 50%
- Vector Weight: 50%

Buttons: BALANCED (50/50), KEYWORD HEAVY (70/30), SEMANTIC HEAVY (30/70)

Hybrid Search Results: Found 10 test cases in 1165ms

Stats: 10 in both indexes, 0 BM25 only, 0 vector only

Timing: BM25: 127ms, Vector: 1023ms

Cost: \$0.000091 (7 tokens)

Search Results (10):

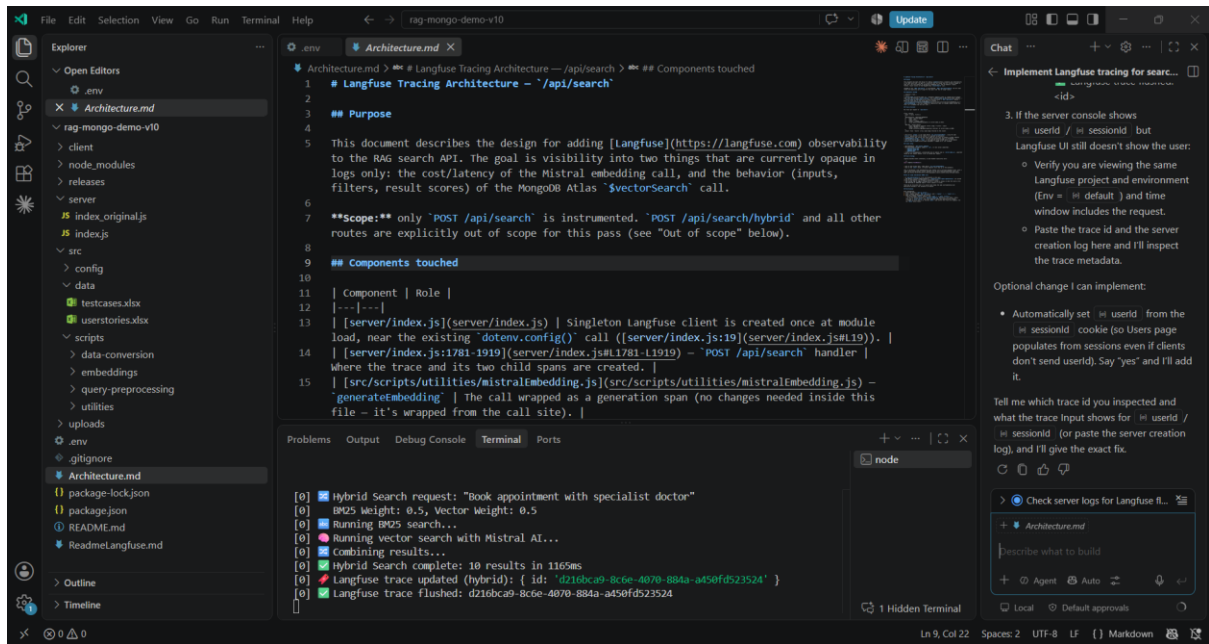
- #1 TC_1143 Hybrid: 94.4% Found in: both

Consultable to book/block appointment

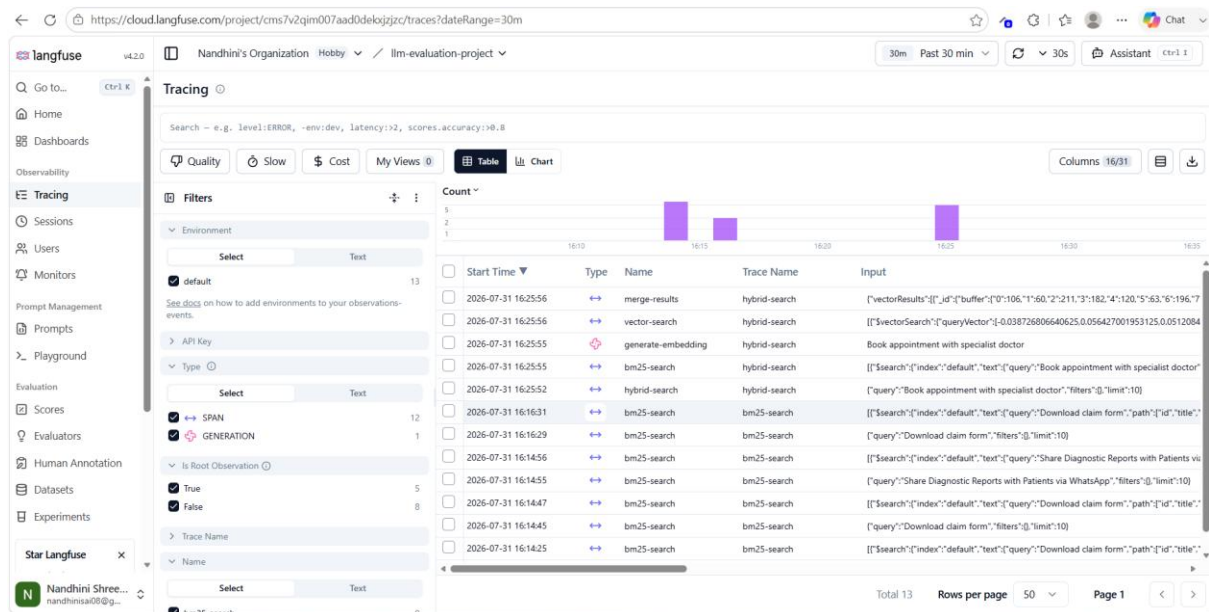
Module: Digital OP **Risk:** Low **Manual**

Description: Verify Consultation able to book appointment/block calendar

Verified the logs for Hybrid Search



Tracing >> Hybrid search



Sessions:

The screenshot shows the Langfuse web interface for the 'Sessions' page. The left sidebar contains navigation links for Home, Dashboards, Observability, Tracing, Sessions (selected), Users, Monitors, Prompt Management, Prompts, Playground, Evaluation, Scores, Evaluators, Human Annotation, Datasets, and Experiments. The main content area displays a table of sessions with columns for ID, Created At, Duration, Environment, Scores, and User IDs. A filters panel on the left allows filtering by Environment (default), Session ID, User IDs, Trace Tags, Metadata, Bookmarked (Not bookmarked), Session Duration, Traces Count, Input Tokens, Output Tokens, and Total Tokens. The table shows four sessions, all with a duration of 2.00s and a score of 1.00. The bottom of the page shows 'Rows per page' set to 50 and 'Page 1 of 1'.

ID	Created At	Duration	Environment	Scores	User IDs
1-1785495352935	2026-07-31 16:25:52	4.00s	default	1.00	1
1-1785494789866	2026-07-31 16:16:29	2.00s	default	1.00	1
1-1785494695405	2026-07-31 16:14:55	1.00s	default	1.00	1
1-1785494685685	2026-07-31 16:14:45	2.00s	default	1.00	1

Sessions for Hybrid Search

The screenshot shows the Langfuse web interface for a specific session, identified by the URL: `https://cloud.langfuse.com/project/cms7v2qim007aad0dekjzjzc/sessions/%3A%3A1-1785495352935?viewId=_langfuse_with_io_`. The session ID is `1-1785495352935`. The interface displays the session details, including the input and output. The input section shows a query: `"Book appointment with specialist doctor"`. The output section shows a successful search result with a score of 1.00. The output details include: `success: true`, `searchType: "hybrid"`, `query: "Book appointment with specialist doctor"`, `filters: empty object`, `weights: 2 items`, `results: [...]`, `count: 10`, `stats: 5 items`, `timing: 3 items`, `cost: 7.000000000000001e-7`, `tokens: 7`, `model: "mistral-embed"`, and `timestamp: "2026-07-31T10:55:56.857Z"`. The bottom of the page shows the session ID and a link to the session details.

Path	Value
query	"Book appointment with specialist doctor"
filters	empty object
limit	10

Path	Value
success	true
searchType	"hybrid"
query	"Book appointment with specialist doctor"
filters	empty object
weights	2 items
results	[...]
count	10
stats	5 items
timing	3 items
cost	7.000000000000001e-7
tokens	7
model	"mistral-embed"
timestamp	"2026-07-31T10:55:56.857Z"

Users

langfuse

v4.2.0

Nandhini's Organization

Hobby

llm-evaluation-project

6h

Past 6 hours

Assistant

Ctrl I

Go to...

Ctrl K

Home

Dashboards

Observability

Tracing

Sessions

Users

Monitors

Prompt Management

Prompts

Playground

Evaluation

Scores

Evaluators

Human Annotation

Datasets

Experiments

Star Langfuse

Nandhini Shree...

nandhinisa08@g...

Users

Search (User ID)

Env default

Filters

User ID	Environment	First Event	Last Event	Total Events	Total Tokens	Total Cost
1	default	31/7/2026, 4:14:45 pm	31/7/2026, 4:25:56 pm	11	7	\$0.00
09564039-d09c-4495-81ee-1f59d47...	default	31/7/2026, 10:41:31 am	31/7/2026, 10:41:35 am	3	12	\$0.00

Rows per page 50Page 1 of 1

langfuse

v4.2.0

Nandhini's Organization

Hobby

llm-evaluation-project

Users

Assistant

Ctrl I

Go to...

Ctrl K

Home

Dashboards

Observability

Tracing

Sessions

Users

Monitors

Prompt Management

Prompts

Playground

Evaluation

Scores

Evaluators

Human Annotation

Datasets

Experiments

Star Langfuse

Nandhini Shree...

nandhinisa08@g...

User 1

Dashboard

Up

Down

Observations: 11

Traces: 4

Total Tokens: 7

Total Cost: \$0.00

Active: 31/7/2026, 4:14:45 pm - 31/7/2026, 4:25:56 pm

Traces

Sessions

Scores

Quality

Slow

Cost

My Views

Search...

IDs / Names

6h

Past 6 hours

30s

Columns 16/31

Download

Filters

Environment

default 11

API Key

Type

SPAN 10

GENERATION 1

Is Root Observation

True 4

False 7

Trace Name

Name

	Start Time	Type	Name	Trace Name	Input
☐	2026-07-31 16:25:56	merge-results	hybrid-search	hybrid-search	[{"vectorResults":[{"id":{"buffer":"0x10611601221131182412056361967112
☐	2026-07-31 16:25:56	vector-search	hybrid-search	hybrid-search	[{"\$vectorSearch":{"queryVector":{"-0.0387268066406250.0564270019531250.051208496
☐	2026-07-31 16:25:55	generate-embedding	hybrid-search	hybrid-search	Book appointment with specialist doctor
☐	2026-07-31 16:25:55	bm25-search	hybrid-search	hybrid-search	[{"search":{"index":"default","text":{"query":"Book appointment with specialist doctor"},"p
☐	2026-07-31 16:25:52	hybrid-search	hybrid-search	hybrid-search	[{"query":"Book appointment with specialist doctor","filters":{"limit":10}
☐	2026-07-31 16:16:31	bm25-search	bm25-search	bm25-search	[{"search":{"index":"default","text":{"query":"Download claim form"},"path":{"id","de
☐	2026-07-31 16:16:29	bm25-search	bm25-search	bm25-search	[{"query":"Download claim form","filters":{"limit":10}
☐	2026-07-31 16:14:56	bm25-search	bm25-search	bm25-search	[{"search":{"index":"default","text":{"query":"Share Diagnostic Reports with Patients via W
☐	2026-07-31 16:14:55	bm25-search	bm25-search	bm25-search	[{"query":"Share Diagnostic Reports with Patients via WhatsApp","filters":{"limit":10}
☐	2026-07-31 16:14:47	bm25-search	bm25-search	bm25-search	[{"search":{"index":"default","text":{"query":"Download claim form"},"path":{"id","title","de
☐	2026-07-31 16:14:45	bm25-search	bm25-search	bm25-search	[{"query":"Download claim form","filters":{"limit":10}

Total 11Rows per page 50Page 1